



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination I
Academic Year: 2024-25 (Semester – II)

Class: FY M.Tech.	Subject: PLM: Advanced Concepts	Subject Code: PCC-PL510
Date: 11.02.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	a Explain four-tier architecture.	05	CO2	BT2
	b Difference between sequencing and versioning in TC	02	CO1	BT2
	c What is FMS? Explain it.	03	CO2	BT1, BT2
Q.2	a What is publishing and unpublishing in multisite collaboration? Explain it.	04	CO4	BT1, BT2
	b Differentiate IDSM and ODS.	04	CO4	BT2
	c What is dataset in PLM?	02	CO1	BT1

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Exhibits the use of various elements of PLM.
CO 2	Selects proper architecture of PLM tools for an Industry.
CO 3	Administer for installing, configuring, integrating and efficient management of PLM tools.
CO 4	Select proper methodology for data transfer and implementation in CAD and PLM systems.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
In-Semester Examination I
Academic Year: 2024-25 (Semester -II)

Class: M.Tech.	Subject: Machine Learning	Subject Code: PEC-PL-518
Date: 13.02.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. This paper contains one page.

Q.N.		Marks	CO	BT
Q.1	Discuss the various types of Machine Learning, providing clear explanations for each. Support your answer with practical, real-world examples that illustrate how each type is applied in different domains or industries.	6	CO 1	BT 1-2
Q.2	What are the key differences between Batch Learning and Online Learning in machine learning? Explain with suitable examples.	4	CO 1	BT 1-2
Q.3	What are the major challenges encountered in developing and deploying machine learning models?	4	CO 1	BT 1-2
Q.4	Explain the key steps of the Machine Learning Development Life Cycle (MLDLC) and discuss the significance of each stage in developing an efficient machine learning model.	6	CO 1-3	BT 1-3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Appreciate the underlying mathematical relationships within and across Machine Learning algorithms and the paradigms of supervised and un-supervised learning.
CO 2	Understand the concept behind neural networks for learning non-linear functions
CO 3	Design and implement various machine learning algorithms in a range of real-world applications.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
ISE-I Examination (Semester - II)
Academic Year: 2024-25

Class: FY-M.Tech.
(Common to All)

Subject: Project Management

Subject Code: AUD-901

Date: 10/02/2025

Time: 11:00 am to 11:45 am

Max. Marks: 20

Note:

- i. Non-programmable calculator is permitted.
- ii. Figures to right indicate full marks.
- iii. This paper contains two pages

Q.No.	Question	Marks	CO	BT																																			
Q1	a Explain the various stages in Project Life Cycle.	05	CO1	BT1																																			
	b Explain the classification of projects on various bases. (any 3)	05	CO1	BT2																																			
Q2	a Explain the various organization structure used in managing projects.	05	CO2	BT1																																			
	Define: 'Tail & Head event' and 'Dummy Activity'. Draw a network for House Construction Project. The sequence of activities and their Predecessor activities is given in table below:																																						
	<table><tr><th>Name of the Activity</th><th>Start & Finish events</th><th>Description of Activity</th><th>Predecessor Activity</th><th>Duration (in Days)</th></tr><tr><td>A</td><td>(1, 2)</td><td>Prepare the house plan</td><td>--</td><td>4</td></tr><tr><td>B</td><td>(2, 3)</td><td>Construct the house</td><td>A</td><td>58</td></tr><tr><td>C</td><td>(3, 4)</td><td>Fix the door/ windows</td><td>B</td><td>2</td></tr><tr><td>D</td><td>(3, 5)</td><td>Wiring the house</td><td>B</td><td>2</td></tr><tr><td>E</td><td>(4, 6)</td><td>Paint the house</td><td>C</td><td>1</td></tr><tr><td>F</td><td>(5, 6)</td><td>Polish the doors/ windows</td><td>D</td><td>1</td></tr></table>	Name of the Activity	Start & Finish events	Description of Activity	Predecessor Activity	Duration (in Days)	A	(1, 2)	Prepare the house plan	--	4	B	(2, 3)	Construct the house	A	58	C	(3, 4)	Fix the door/ windows	B	2	D	(3, 5)	Wiring the house	B	2	E	(4, 6)	Paint the house	C	1	F	(5, 6)	Polish the doors/ windows	D	1	05	CO2	BT2
Name of the Activity	Start & Finish events	Description of Activity	Predecessor Activity	Duration (in Days)																																			
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E	(4, 6)	Paint the house	C	1																																			
F	(5, 6)	Polish the doors/ windows	D	1																																			

== ALL THE BEST ==



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
ISE 1 Examination
Academic Year: 2024-25 (Semester – II)

Class: M.Tech (II Sem)	Subject: Internet of Things (Elective-IV)	Subject Code: PEC-PL-519
Date: 12.02.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed.
- Figures to the right indicate full marks.
- Indicate the Question Numbers while answering each question correctly.

Q.N.	Solve any TWO out of THREE sub-questions in Each .	Marks	CO	BT
Q.1	a Explain the concept of IoT, its characteristics, advantages and disadvantages? How it enables smart applications like smart cities and healthcare systems.	5	CO1	BT1
	b Describe the architecture of IoT in detail, highlighting the roles of the perception, network, processing, application, and business layers.	5	CO1	BT1
	c Elaborate "IoT can transform Businesses".	5	CO1	BT1
Q.2	a Explain the architecture and key components of a sensor network. How do energy efficiency and network scalability impact sensor network performance?	5	CO2	BT1
	b Differentiate M2M communication with IoT in terms of connectivity, protocols, and data processing. Provide real-world examples to support your answer. (03 + 02 = 05 Marks)	5	CO2	BT4
	c Explain the structure of an Arduino sketch (Architecture and Pin Diagram). Write a simple Arduino program to read temperature data from a sensor and display it on the LCD.	5	CO2	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To introduce the concept of Internet of Things and its horizon in terms of Application.
CO 2	To make aware the importance of sensor network and interoperability and Open Source Physical computing platform Arduino.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
End Semester Examination I
Academic Year: 2024-25 (Semester – II)

Class: FY M.Tech.	Subject: Web and Network Technologies	Subject Code: PCC-PL-511
Date: 14.02.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry the usual meaning.

Q.N.		Marks	CO	BT
Q.1	a. Explain the physical layer in OSI model	3	CO1	BT 1-2
	b. Discuss the working of data link layer in OSI model.	4	CO1	BT 1-2
	c. Discuss the working of transport ^{Network} layer in OSI model.	3	CO1	BT 1-2
Q.2	a. Explain the working of Repeaters.	5	CO1	BT 1-2
	b. Discuss the working of Bridge and its functions ^{application} .	4	CO1	BT 1-2
	c. Explain the working of Routers.	4	CO2	BT 1-2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Select proper web and internet network.
CO 2	Gain knowledge of Networking Protocols, Internet working concepts, History and Architecture.
CO 3	Understand working of WWW, HTTP, TELENET, JavaScript, and Ajax, PHP/MySQL, ASP.Net.
CO 4	Gain knowledge of Java Web Technologies.
CO 5	Select proper Web Security and Network Security.
CO 6	Understand the concept of XML and Cloud Computing.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
ISE-I Examination (Semester - II)
Academic Year: 2024-25

Class: FY-M.Tech.
(Common to All)

Subject: Project Management

Subject Code: AUD-901

Date: 18/03/2025

Time: 11:00 am to 11:45 am

Max. Marks: 20

Note:

- i. Non-programmable calculator is permitted.
- ii. Figures to right indicate full marks.
- iii. This paper contains two pages

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Q.No.	Question									Marks	CO	BT	
Q1	a	State and Explain Different Techniques of Project Management.									07	CO2	BT2
	b	Define: Optimistic, Pessimistic, and most likely time estimate.									03	CO2	BT1
Q2	a	The data for a project network is as follows, draw the A-O-N network, calculate the project duration, and indicate the critical path. Prepare table showing EST, EST, LST, LFT and total float the various activities of the Project.									10	CO2	BT3
		Activity	P	Q	R	S	T	U	V	W			
		Immediate Predecessor	-	P	P	Q	Q	R	S	T, U			
		Duration (Weeks)	3	4	6	6	4	4	6	8			

== ALL THE BEST ==

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